

The Course

...an (experimental) **SAR training course**

Course No 8: 4th of November 2024 – 7th of February 2025

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

Course Organisation: ESA, DLR, EEBIOMASS



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Motivation

A rapidly growing (scientific and commercial) community with limited or no SAR background uses (or intends to use) multi-parameter SAR data:

... a new generation of multi-parameter SAR missions with open data policy is already operational in orbit or about to be launched;

... synergies between SAR and optical RS data (or products) developed in recent years confront the "optical community" with the use of multi-parameter SAR data.

The interpretation of SAR data is (compared to optical data) less intuitive, while the available literature is often "too mathematical", aiming at a more engineering oriented audience, or oversimplified.

Understanding the information content of SAR data and the basic SAR processing principles/algorithms is neither as difficult nor as complex as it might appear. Anyone willing to invest a reasonable amount of time can achieve both.

The Pol-InSAR course is an online hands-on course that aims to develop the understanding on the information content of multi-parameter SAR data and to eliminate the respect for SAR data processing in an interactive way, without assuming any prior / background knowledge.

The course is run on ESA's Multi-Mission Algorithm and Analysis Platform (ASCENT), so there are no hardware or software requirements for participants.



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Course Outline: 5 SAR (2D and 3D) Techniques in 11 Units

Synthetic Aperture Radar - SAR

1. **Focusing in Range:** Spatial resolution in Range, Matched filter, 1D and 2D focusing, ...
2. **Focusing in Azimuth:** Spatial resolution in Azimuth, Synthetic aperture, 1D and 2D focusing, ...

SAR Interferometry - InSAR

1. **Interferogram formation:** Image co-registration, flat earth removal, geometric interpretation, ...
2. **Interferometric coherence:** InSAR decorrelation, volume decorrelation, forest height inversion, ...
3. **Differential Interferometry:** D-InSAR principles, LOS deformation, propagation effects, ...

SAR Polarimetry - PolSAR

1. **Scattering matrix:** Polarimetric SAR, physical interpretation of scattering mechanisms, ...
2. **Covariance matrix:** Distributed scatterers, depolarization, polarimetric eigen-decomposition, ...

Polarimetric SAR Interferometry - Pol-InSAR

1. **The Pol-InSAR space:** Pol-InSAR observation space, interpretation and representation, ...
2. **Model based Pol-InSAR inversion:** forest height inversion, dual-pol implementation, ...

SAR Tomography - TomoSAR

1. **3D Focusing:** Vertical aperture formation, 3D resolution, 3D focusing, ...
2. **3D Refeectivity:** Reconstrucion algorithms, polarimetric extension, interpretation, ...

Each Unit = A Week with 3 Sessions !

Mon

Theoretical Background (~60min) + Break + Introduction of Exercise(s) (~60min)

Tue

Wed

Check Point: Discussion of problems & results, questions & answers (~60min)

Thu

Fri

Closing Session: Discussion of results, questions & answers (~90 + min)

Sat

Sun

- ▶ Zoom Link: <https://zoom.us/j/98253602750?pwd=U0xBOXUrSFY2NHZ6bzBuU3oyWWUxdz09>
- ▶ The exercises are performed on ESA's Mission Algorithm and Analysis Platform (ASCENT)
- ▶ The sessions are recorded and the recording is distributed after each session

Proposed Timeline

Synthetic Aperture Radar - SAR	November 2024	December 2024	January 2025	February 2025
1. Focusing in Range	1 Fr	1 Su	1 We New Year's Day	1 Sa
2. Focusing in Azimuth	2 Sa	2 Mo	2 Th	2 Su
	3 Su	3 Tu	3 Fr	3 Mo
	4 Mo	4 We	4 Sa	4 Tu
	5 Tu	5 Th	5 Su	5 We
	6 We	6 Fr	6 Mo	6 Th
	7 Th	7 Sa	7 Tu	7 Fr
	8 Fr	8 Su	8 We	8 Sa
	9 Sa	9 Mo	9 Th	9 Su
	10 Su	10 Tu	10 Fr	10 Mo
	11 Mo	11 We	11 Sa	11 Tu
	12 Tu	12 Th	12 Su	12 We
	13 We	13 Fr	13 Mo	13 Th
	14 Th	14 Sa	14 Tu	14 Fr
	15 Fr	15 Su	15 We	15 Sa
	16 Sa	16 Mo	16 Th	16 Su
	17 Su	17 Tu	17 Fr	
	18 Mo	18 We	18 Sa	18 Tu
	19 Tu	19 Th	19 Su	19 We
	20 We	20 Fr	20 Mo	20 Th
	21 Th	21 Sa	21 Tu	21 Fr
	22 Fr	22 Su	22 We	22 Sa
	23 Sa	23 Mo	23 Th	23 Su
	24 Su	24 Tu	24 Fr	24 Mo
	25 Mo	25 We Christmas Day	25 Sa	25 Tu
	26 Tu	26 Th	26 Su	26 We
	27 We	27 Fr	27 Mo	27 Th
	28 Th	28 Sa	28 Tu	28 Fr
	29 Fr	29 Su	29 We	
	30 Sa	30 Mo	30 Th	
		31 Tu	31 Fr	

CET - 9

CET - 5

CET

CET + 4.30

CEST + 7



The Time Challenge



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CET - 9

03:00

03:30

04:00

CET - 4

08:00

08:30

09:00

CET

12:00

12:30

13:00

CET + 4.30

16:30

17:00

17:30

CET + 7

22:00

22:30

23:00

The Time Challenge



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Each Unit = 3 Sessions in a Week !

Mon

Theoretical Background (~60min) + Break + Introduction of ... (~60min)

12:00 – 14:00 CET

Tue

Wed

Check Point: Discussion of problems & results, questions & answers (~60min)

12:00 – 13:00 CET

Thu

Fri

Closing Session: Discussion of results, questions & answers (~60min)

12:00 – 14:00 CET

Sat

Sun

- ▶ Zoom Link: <https://zoom.us/j/98253602750?pwd=U0xBOXUrSFY2NHZ6bzBuU3oyWWUxdz09>
- ▶ The exercises are performed on ESA's Mission Algorithm and Analysis Platform (MAAP)
- ▶ The sessions are recorded and the recording is distributed after each session

ESA's Mission Algorithm and Analysis Plattform (ASCENT)

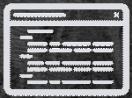
The course is hosted by ESA's Mission Algorithm and Analysis Plattform (MAAP) that is a virtual open and collaborative environment that:



Enables researchers to easily discover, process, visualize and analyze large volumes of data.



Provides tools and infrastructures to bring data into the same coordinate reference frame to enable comparison, analysis and evaluation.



Provides a version-controlled science algorithm development environment that supports tools, co-located data, and processing resources.



Addresses intellectual property and sharing issues related to collaborative algorithm development and sharing of data and algorithms.



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Overview of Eclipse Che - JupyterLab Resources on the MAAP

Main user interface: JupyterLab



Main programming language: Python



Additional support for: R and Julia



Main packages: NumPy, SciPy, GDAL, Scikit-Learn, Dask, Xarray



Available resources per user:



RAM: 10GB



Storage: 30 GB

There are no requirements on any hardware / software components for the user



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